READING BETWEEN THE LINES

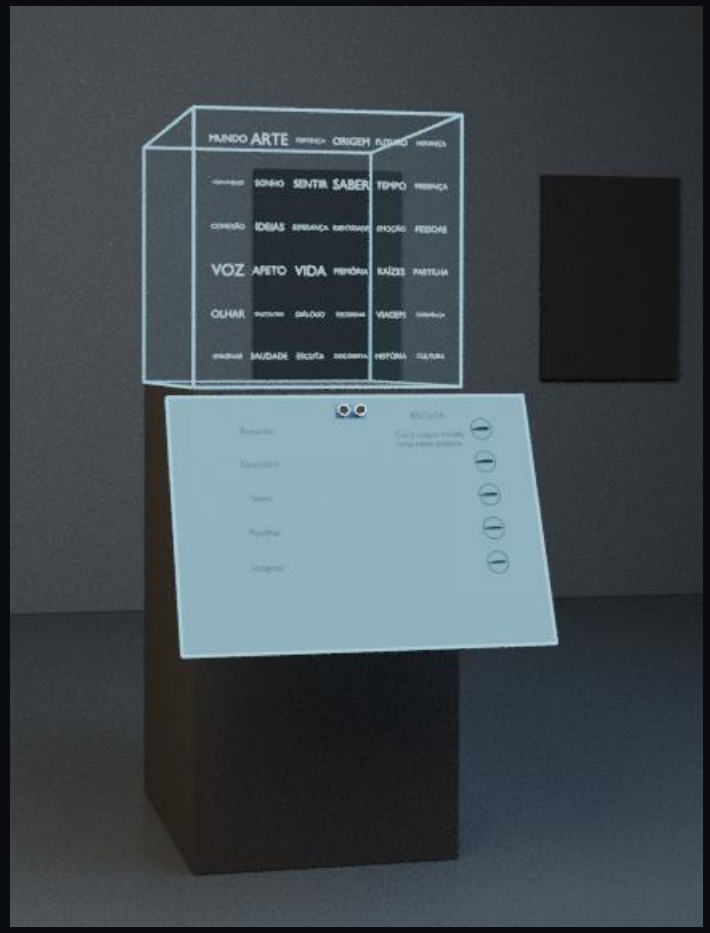
MILESTONE 1 – PROJECT 1 – MASTERS IN CREATIVE COMPUTER AND ARTIFICIAL INTELLIGENCE – 2026/2027

RUI DURÃO

TYPE DATASETS
by Rudolfo Quintas



READING BETWEEN THE LINES
by Rui Durão



INÍCIO FEDERAÇÃO SINDICATOS NOTÍCIAS

Início > Notícias > PISA 2025: FNE alerta que Portugal está a perder terreno nas aprendizagens

PISA 2025: FNE alerta que Portug
aprendizagens

Diário de

ria Últimas Opinião Vídeos DN Sport Barómetro DN / Axima

Foto: Pedro Granadeiro

Sociedade

nos portugueses com "resulta
óricos, mas no sentido negat
A

rio da Educação descarta responsabilidades polític
mento das aprendizagens, mas diz-se responsável
es. Alunos portugueses recuaram ao início do sécul

RTP notícias_ PAÍS EUROPA MUNDO CULTURA OPINIÃO VÍDEOS ÁUDIOS

PAÍS

Relatório PISA. Alunos portugueses mostram estar a perder
conhecimentos e competências

P 1.ª HORA NOVO EDUCAÇÃO SAÚDE JUSTIÇA MEDIA FLORESTAS PÚBLICO NA ESCOLA MAIS

EXCLUSIVO PISA 2025

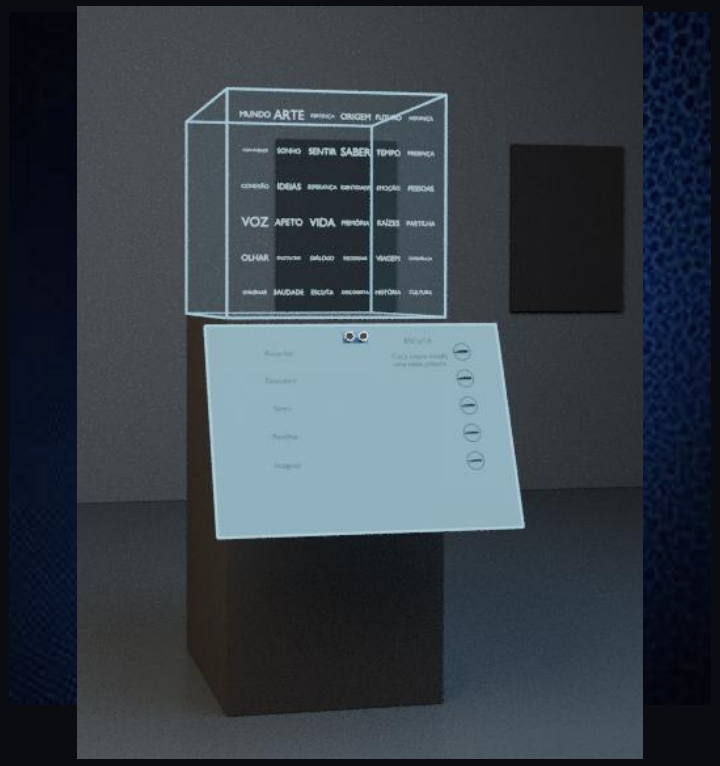
Portugal regride 20 anos a Matemática e Leitura, Alentejo e Setúbal no fim da tabela

Há mais alunos portugueses que não conseguem chegar ao nível mínimo necessário para "participar eficazmente

NEWS CLIPPINGS AS MOTIVATOR

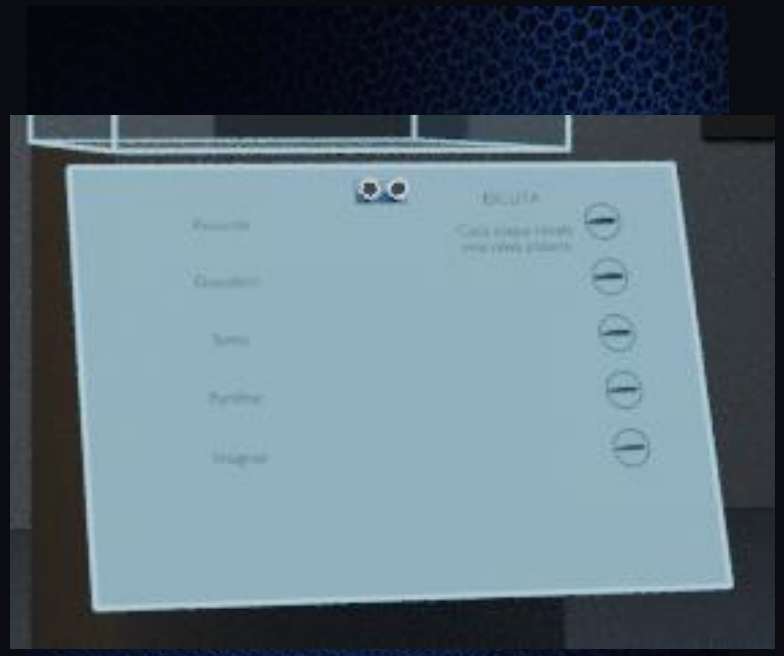
Several news about bad results of portuguese students was a motivator for Education Topic of READING BETWEEN THE LINES installation

READING BETWEEN THE LINES CONCEPT



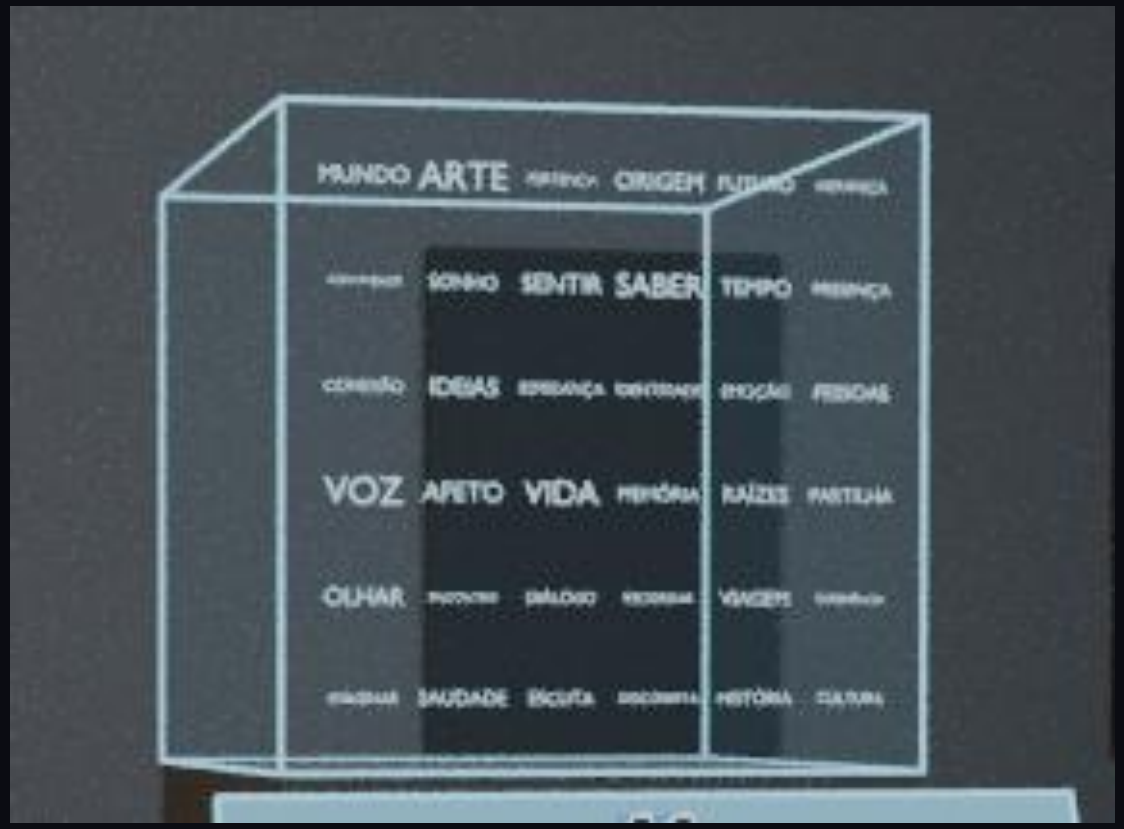
- Visitors explore their sentiments over education in Portugal, with data from PISA study.
- While idle, the instalation narrate using texto-to-speech and word cloud, related to the state of education based on PISA data, from 2009 to 2025.
- Visitors get closer, interrupt the narration and express their feelings over the data exposed.
- Additional toggle will be added for future forecasting

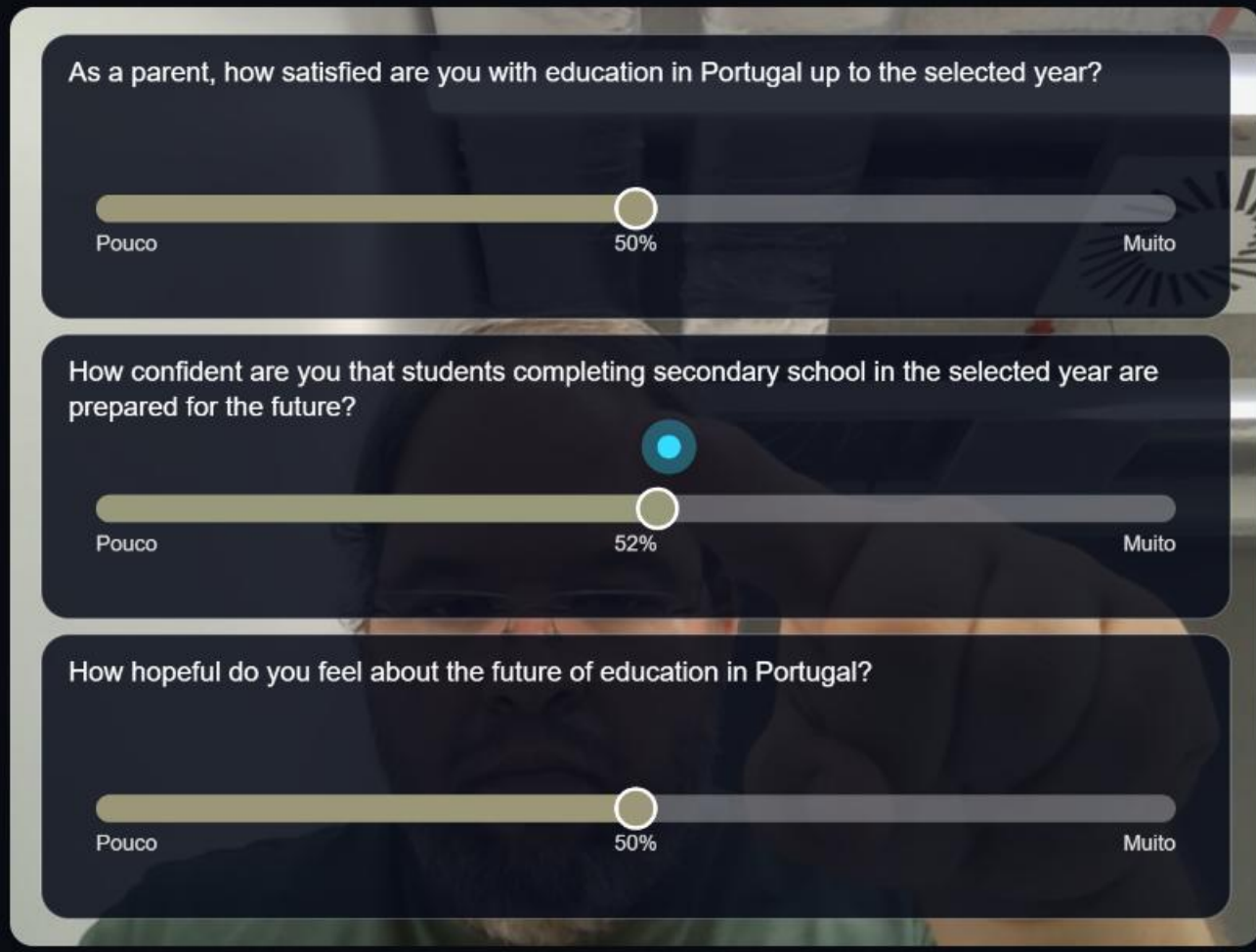
SENTIMENT GATHERING

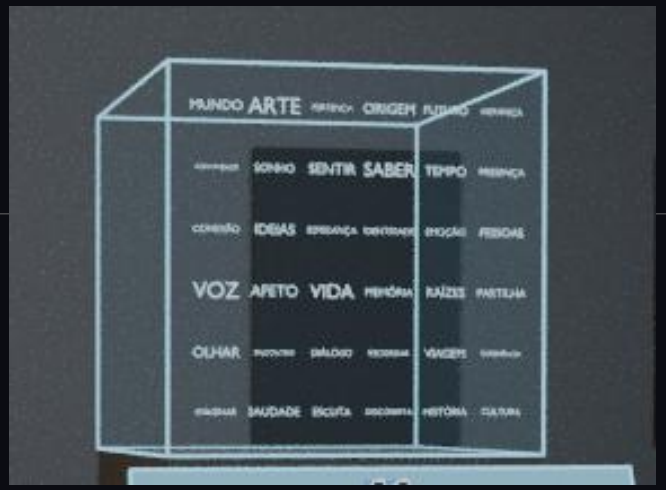


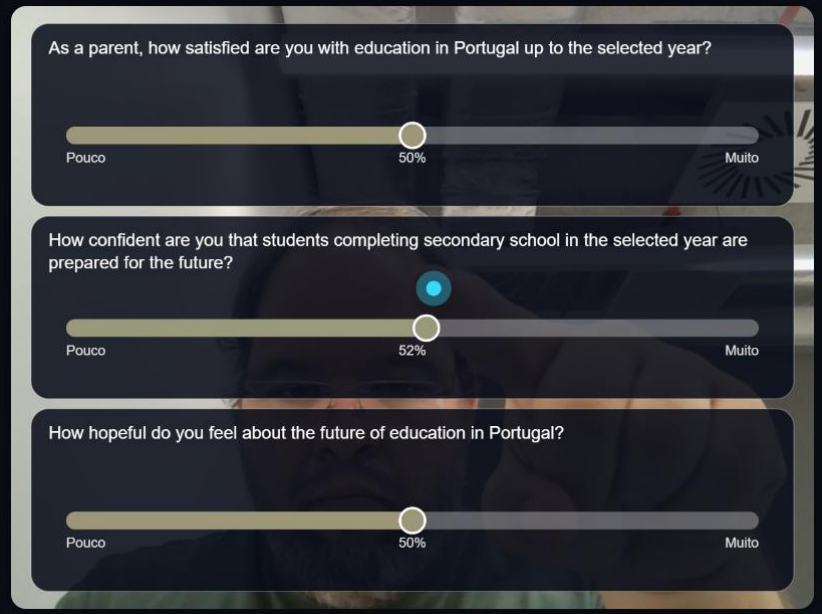
- Acrylic transparente plaque, with engraved questions, for exemple:
  - “As a parent, how satisfied are you with education in Portugal up to the selected year?”
  - “How confident are you that students completing secondary school in the selected year are prepared for the future?”
  - Additional questions exploring educational opportunities, learning experiences, and expectations.
- And a pressure sensor to measure sentiment

“MINORITY REPORT” STYLE ALTERNATIVE INTERACTION

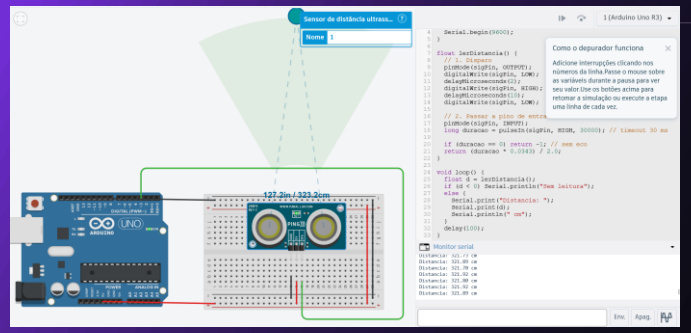


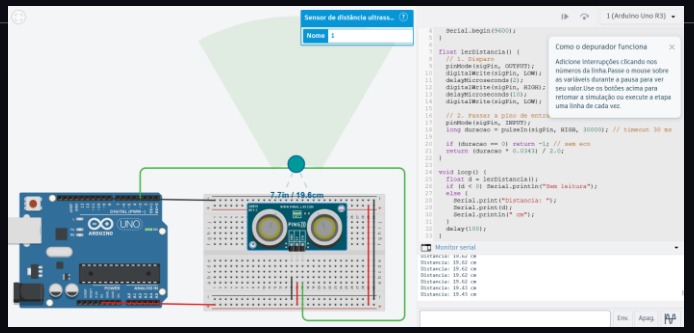


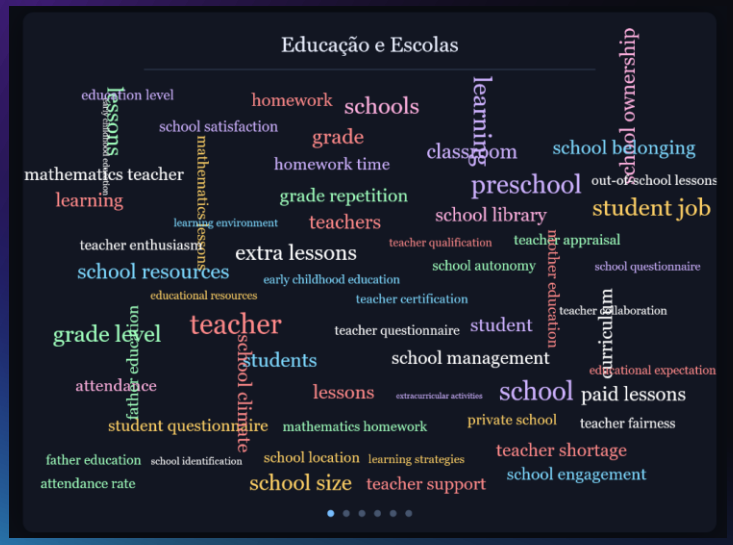


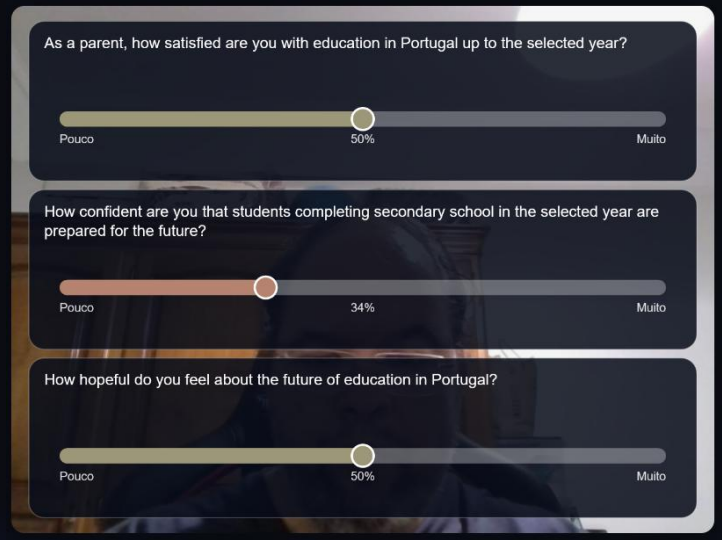


- Alternative interaction, not physical, with finger tracking, visitor can express their sentimento over the statements
- This view is toggle when a person get closer to the ultrasonic sensor, stoping narration

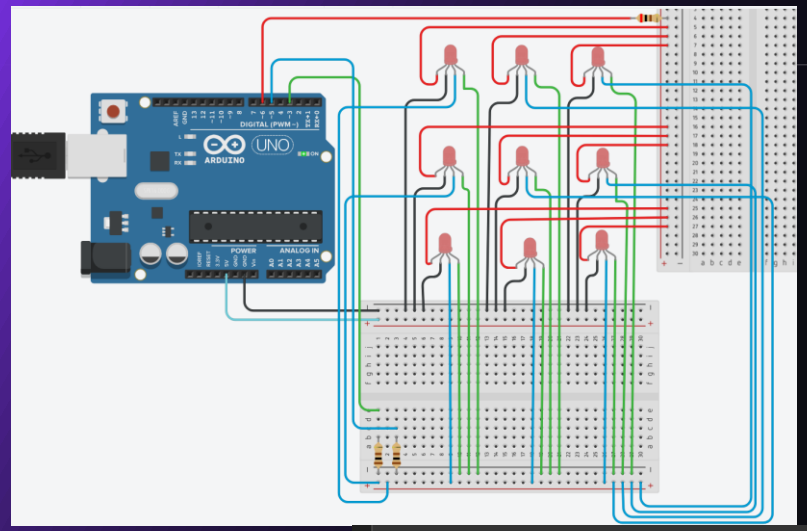


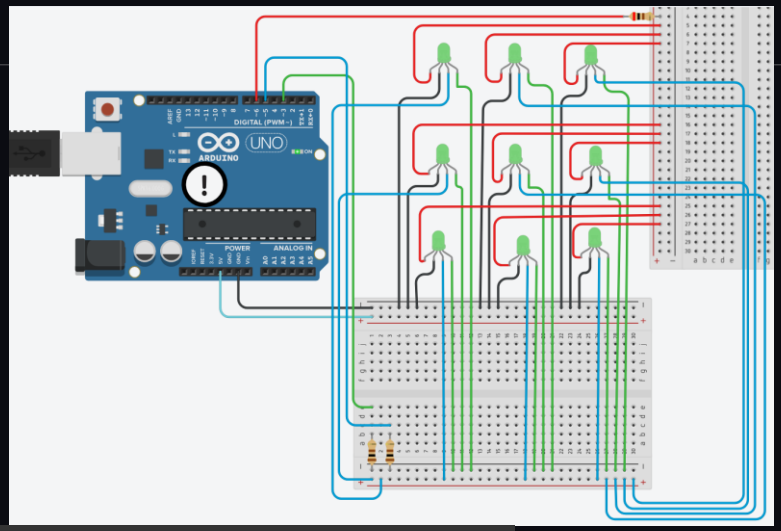


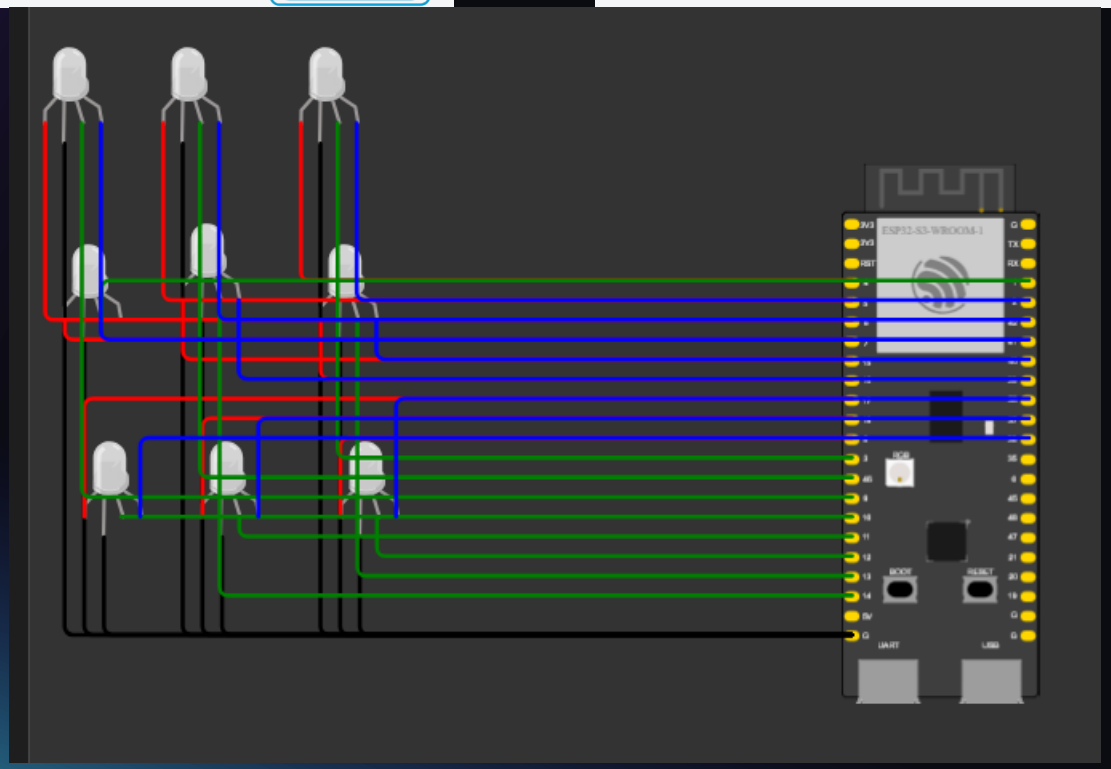

Educação e Escolas
education level
homework
schools
school satisfaction
grade
learning
classroom
school belonging
school ownership
mathematics teacher
homework time
preschool
out-of-school lessons
learning
grade repetition
student job
learning environment
teachers
school library
teacher enthusiasm
teacher qualification
teacher appraisal
extra lessons
school resources
school autonomy
school questionnaire
early childhood education
educational resources
teacher certification
teacher collaboration
grade level
teacher
teacher questionnaire
student
curriculum
students
school management
educational expectation
attendance
lessons
school
paid lessons
student questionnaire
mathematics homework
private school
teacher fairness
father education
school identification
school location
learning strategies
teacher shortage
attendance rate
school size
teacher support
school engagement


As a parent, how satisfied are you with education in Portugal up to the selected year?
Pouco
50%
Muito
How confident are you that students completing secondary school in the selected year are prepared for the future?
Pouco
34%
Muito
How hopeful do you feel about the future of education in Portugal?
Pouco
50%
Muito

DEMO

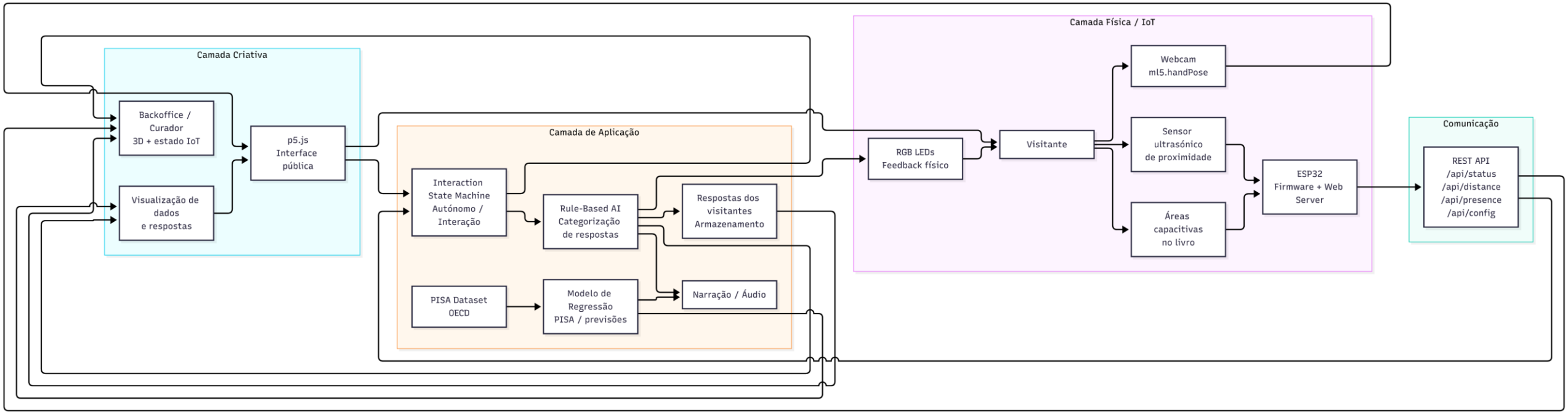




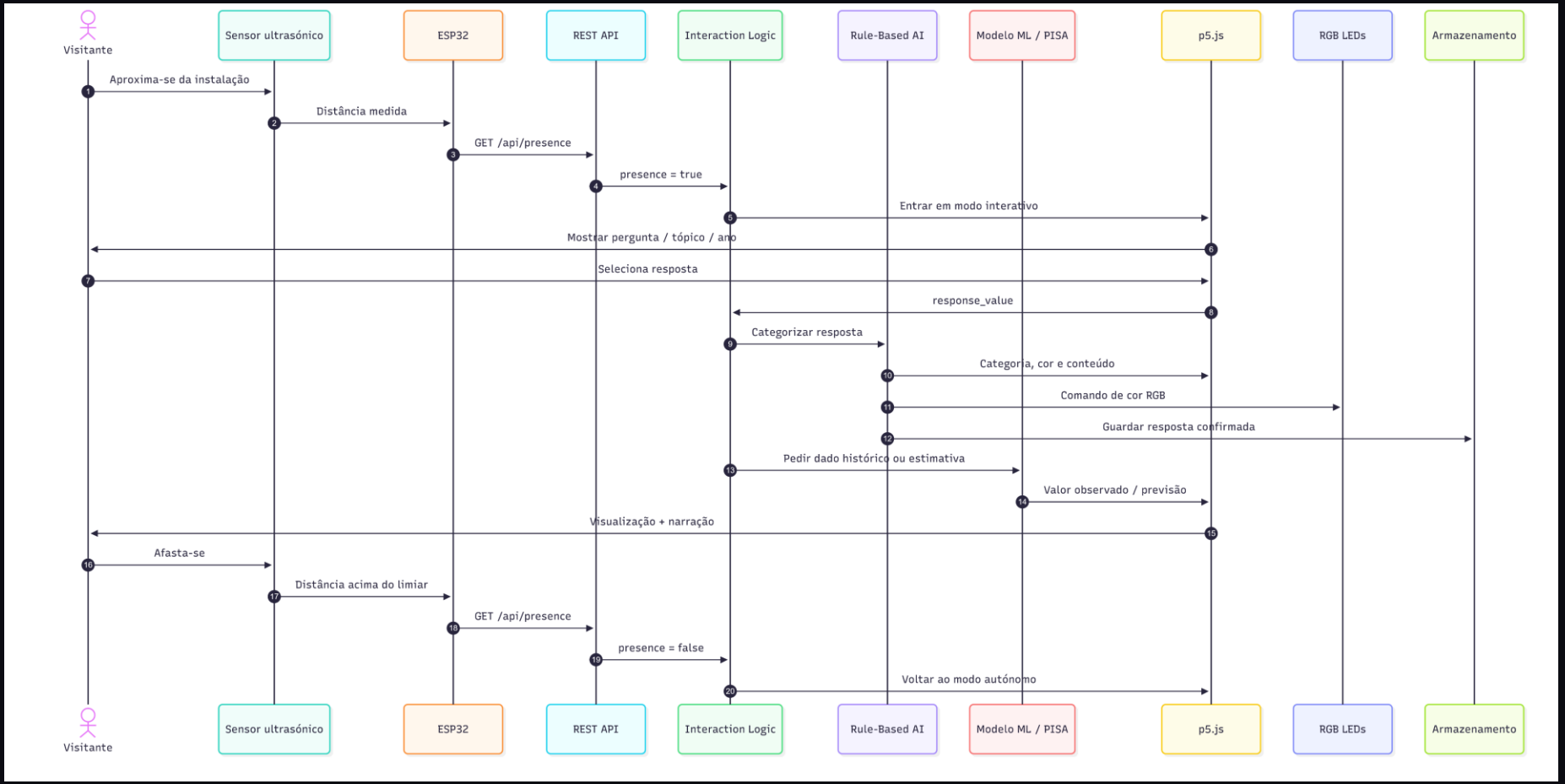


DEMO

DATA AND COMPONENTS FLOW



USE CASE


Visitante
Sensor ultrasónico
ESP32
REST API
Interaction Logic
Rule-Based AI
Modelo ML / PISA
p5.js
RGB LEDs
Armazenamento
1 Aproxima-se da instalação
2 Distância medida
3 GET /api/presence
4 presence = true
5 Entrar em modo interativo
6 Mostrar pergunta / tópico / ano
7 Seleciona resposta
8 response_value
9 Categorizar resposta
10 Categoria, cor e conteúdo
11 Comando de cor RGB
12 Guardar resposta confirmada
13 Pedir dado histórico ou estimativa
14 Valor observado / previsão
15 Visualização + narração
16 Afasta-se
17 Distância acima do limiar
18 GET /api/presence
19 presence = false
20 Voltar ao modo autónomo
Visitante
Sensor ultrasónico
ESP32
REST API
Interaction Logic
Rule-Based AI
Modelo ML / PISA
p5.js
RGB LEDs
Armazenamento

ACTION!! (VIDEO DEMO)

ROLES OF COURSES

Introduction to AI

- Provide knowledge base to use pre-trained hand-tracking model to detect deliberate gestures and a rule-based system to translate those actions, along with selected topics and response values, into predefined opinion categories and coordinated visual, audio, and narrative feedback.

Applied Machine Learning

- Provide base to use supervised multiple regression on OECD PISA-derived educational data to examine relationships between selected explanatory variables and a defined educational outcome, with its estimates—kept clearly separate from visitors' perceptions—informing the installation's p5.js visual and narrative content.

ROLES OF COURSES

Creative Programming

- Translate educational data, visitor responses, and interaction states into a coherent p5.js-based visual experience—using ml5.js for machine-learning-supported interaction—that connects the acrylic plaque, illuminated cube, and screen content through typography, colour, animation, and data visualisation, while also providing a curator/backoffice interface for monitoring sensors, device states, and simulated interactions.

Physical Computing and IoT

- Connect visitors, the physical installation, and the digital application by using an ESP32 to acquire ultrasonic proximity and capacitive touch inputs, transmit validated sensor data through a Wi-Fi REST API to the p5.js interaction logic, and control RGB lighting feedback, with further development required to complete touch input, lighting integration, and live end-to-end operation.